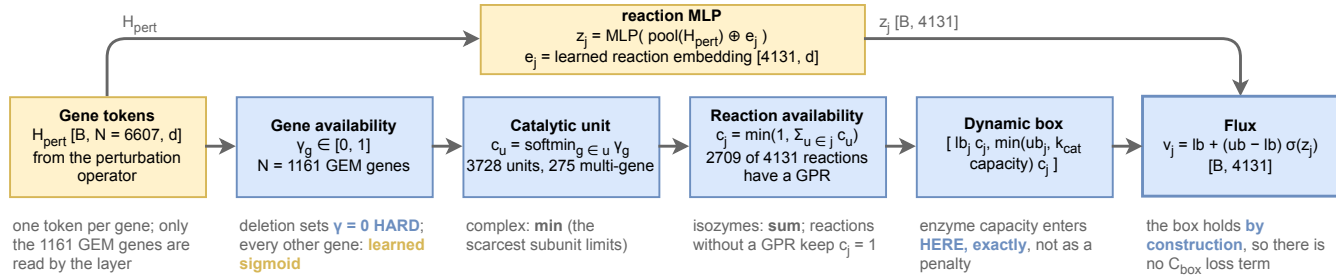
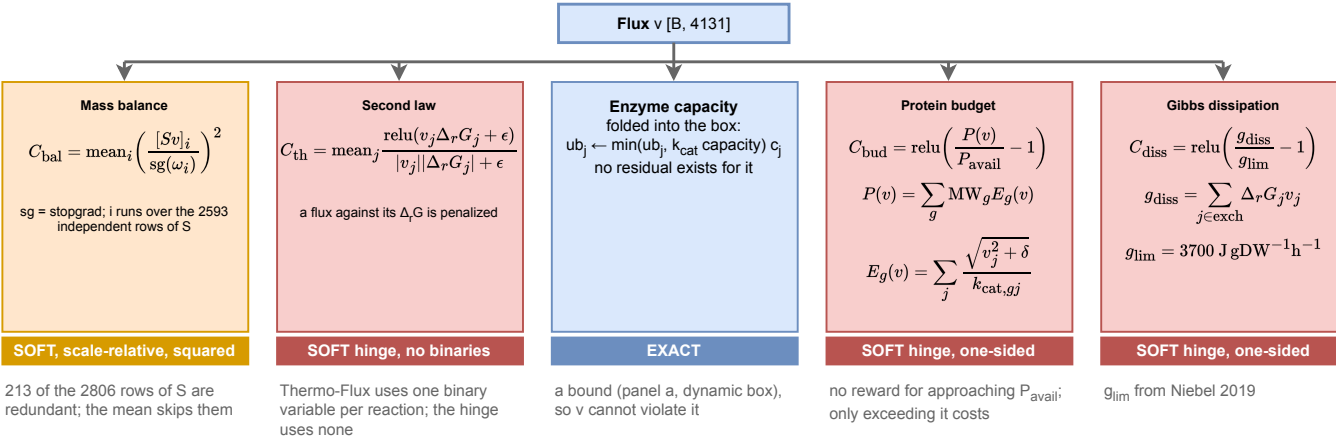


A differentiable enzyme-constrained thermodynamic flux layer

a Availability to box to flux. A deletion acts on the bounds, so the box is exact and enzyme capacity never needs a penalty.



b The constraint residuals hanging off v. One is exact by construction; the other four are soft, and none needs a binary variable.



Thermodynamic decomposition of $\Delta_r G_j$, the driving force used by the second law and by dissipation

$$\Delta_r G_j = \Delta_r G_j^{\text{to}} + RT \sum_i S_{ij} \ln c_i + \Delta_r G_j^t$$

tabulated: $\Delta_r G^0$ known for 85.1% of metabolites

learned $\ln c_i$, bounded to 0.1 μM to 10 mM; anchored where a measurement exists

transport term: recoverable for 441 transport reactions from the GEM's table; off by default

What rests on data, and how much of it is there

S	100%
$\Delta_r G^0$	85.1% of metabolites
MW	100%
k_{cat}	4.0%: 148 of 3728 catalytic units resolved from the Open Enzyme Database
conc. anchors	6.8% of metabolites

exact / structural

soft, squared, scale-relative

soft, one-sided hinge

data-backed quantity

learned

default, or a gap in the data